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MAPPING THE ITALIAN THIRD SECTOR: INSIGNTS FROM WEB-AUGMENTED ADMINISTRATIVE DATA

THE SIGNIFICANCE OF THE ITALIAN THIRD SECTOR

- Today, the **Third Sector** (part of the Social Economy) plays a significant role in the **creation and distribution of national well-being** in most advanced countries.
- In **Europe**, the **Social Economy** accounts for more than **4.3 million entities** and employs more than **11.5 million people**, with **912 billion euros** of turnover (European Commission, 2024). These organizations operate across diverse economic areas, including social assistance, healthcare, social cohesion, sports, and cultural activities.
- In Italy, the most recent comprehensive picture of the sector is provided by the Permanent Census of Nonprofit Institutions by Istat.

360,061	919,431	4,616,915
ORGANIZATIONS	EMPLOYEES	VOLUNTEERS (2021)

in Italy, in 2022

(Istat, Permanent Census of Non-Profit Institutions, 2024)

THE 2017 REFORM AND THE THIRD SECTOR SINGLE NATIONAL REGISTER (RUNTS)

- The d.lgs. 117/2017 widely reviewed the legislation about the Italian Third Sector.
- Among other things, this reform introduced the idea of Third Sector Organizations (TSOs) that benefit from fiscal and administrative advantages and simplifications.
- To qualify as TSO, an organization:
  - Must carry out an economic activity of “general interest”, as defined by the law.
  - Must act for civic, solidarity, and social utility purposes.
  - Must enroll in the Third Sector Single National Register (RUNTS).

THE THIRD SECTOR SINGLE NATIONAL REGISTER (RUNTS)

- The RUNTS lists about 136,170 TSOs, today.
- The register represents a subset (35%) of the non-profit organizations covered by the Istat's Census.*
- The register is publicly available, since the beginning of 2024, via a web platform (https://servizi.lavoro.gov.it/runts/) on which it is possible to find information such as:
  - The number of associates, volunteers, and employees of each TSO.
  - The economic activity (ICNPO classification) carried out by the organization.
  - Legal documents, such as the balance sheets and social reports, of the TSO.

* Please note that the two sets are not fully homogeneous for at least two reasons. First, to enroll in the RUNTS is not compulsory. Second, some non-profit organizations do not pursue activities of "general interest," like a trade union.

ENRICHING TSO INFORMATION FOR ECONOMIC ANALYSIS

The RUNTS represents a pivotal resource for statistical analysis on the Third Sector. Indeed, it contains a wealth of information on every TSO and can serve as a population register to start other surveys. Nonetheless, the information is self-declared and not validated by the Ministry.

We create an original database by validating and enriching the RUNTS:

- Integration of web-collected information to enhance the dataset:
  - Identification of the website of TSOs registered in the RUNTS, taking snapshots of ETS websites quarterly.
  - Extraction and analysis of textual content
- Data Enrichment using NLP techniques, including LLMs:
  - Classification of TSOs based on their economic activities.
  - Additional dimensions of interest derived from website content (geographical scope of the activity, beneficiaries)

RUNTS Data Collection
TSO web address discovery
TSO web pages collection
TSO economic activity classification

RUNTS DATA COLLECTION

- **Objective:** To build an automated pipeline for collecting and validating data from the RUNTS for scientific research.
- **Data Accessibility:** Being public data, the register should be accessible. But the RUNTS portal only allows manual, individual searches, making large-scale systematic analysis impossible.
  - A "FOIA" request revealed the Ministry views individual queries as sufficient, citing "operational burden" as a reason for not providing bulk machine-readable data.
- **Scraping Strategy:** We developed a scraper that uses an official list of tax codes to simulate human behavior, querying each entity to extract granular details not found in the basic spreadsheet.
- **Technical Challenges:** The web platform lacks an API and consistent HTML metadata; some information is only discernable through page layout and rendering rather than code structure.
- **Data Inconsistency:** Much of the data is provided voluntarily by non-profits in free-form text without input validation, requiring the scraper to collect raw data for ex-post restructuring. Information on personnel and their roles is presented in complex multi-column layouts, requiring the scraper to deduce keys from text content rather than tags.
- **Output:** The final pipeline successfully extracts information for 134,815 organizations (in 2025 Q2), including business profiles, economic activities (ICNPO classified), organizational capacity (workers/volunteers), and governing body structures.

L'ASSOCIAZIONE APS

Repertorio: 99999 **Codice fiscale:** 90909090909

Iscritto il 01/01/2024

Iscritto nella sezione in data 01/01/2024

Dati ente

Sezione ASSOCIAZIONI DI PROMOZIONE SOCIALE

Iscrizione come rete associativa No

Codice fiscale 90909090909

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Atto costitutivo 01/01/2024

Ultimo aggiornamento statutario 01/06/2024

Persona 1

Tipo Persona Fisica

Rappresentante legale Sì

Codice fiscale

ABCABC12Z89X456Y

Nome MARIO

Cognome ROSSI

Stato I

Data di nascita 01/01/1980

Provincia RM

Comune ROMA

Carica 1

Carica LEGALE

RAPPRESENTANTE

Data nomina 01/01/2023

TSO WEB ADDRESS DISCOVERY

- For a small number of TSOs we know their website from the register.
- We use Microsoft Bing to increase the number of websites associated with a TSO. The URLs detected are validated searching for information pieces (tax ID, full address, etc.) on the website.
- We assign a URL to 26,359 organizations (about 20% of the total).
- For each TSO with a (validated) web address, we download its home-page and up to 20 other pages (prioritizing pages like “about”, “contacts”, “mission” and shorter URLs).

MICROSOFT BING SEARCH REPLACEMENT

ALTERNATIVE SEARCH ENGINE

- We could switch to an alternative search engine, like Google Search APIs.
- This solution is the closer to our current implementation, but it requires some work to adapt our pipeline and it will be more costly than the current service.

GROUNDING WITH BING SEARCH

- We could use the alternative proposed by Microsoft Azure: Grounding with Bing Search.
- This service integrates directly with LLMs and enriches the AI agent with skills to search the Web for an answer, if it is not in the training set.
- Likely, this solution is cheaper/faster; but it is a black box outside out control.

TSO ECONOMIC ACTIVITY CLASSIFICATION

- We use the text extracted from the website of a TSO to classify the economic activity carried out by the organization according to the ICNPO (International Classification of Non-Profit Organizations) of the United Nations (similar to the NACE classification).
- Among some tested alternatives, we chose:
  - To use OpenAI's o4-mini.
  - To prioritize the most relevant web pages (like "about" or "mission").
  - To extract some key information (like the mission, goals, and beneficiaries of TSOs' activity) about the activity of each TSO from the web page first.
  - To classify the economic activity of the TSO based on such key information.
- Out of the 26,359 organizations we collected the website of, we assigned at least an ICNPO class to 19,641 (74,5%) of them (about 15% of the total).

TSO ECON. ACTIVITY: FIRST PROMPT

You are an expert in analyzing non-profit organizations. Your task is to extract important information from the text provided, focusing on (1) the mission of the organization; (2) its goals; (3) areas of intervention; (4) activities performed; (5) beneficiaries; (6) and main geographical area of activity.

TSO ECON. ACTIVITY: SECOND PROMPT

You are an expert in classifying non-profit organizations according to the ICNPO standard. Please, analyze the information previously extracted to assign the most appropriate ICNPO categories among the following:

1000 CULTURE AND RECREATION
- 1100 Culture and Arts
- 1200 Sports
[…]

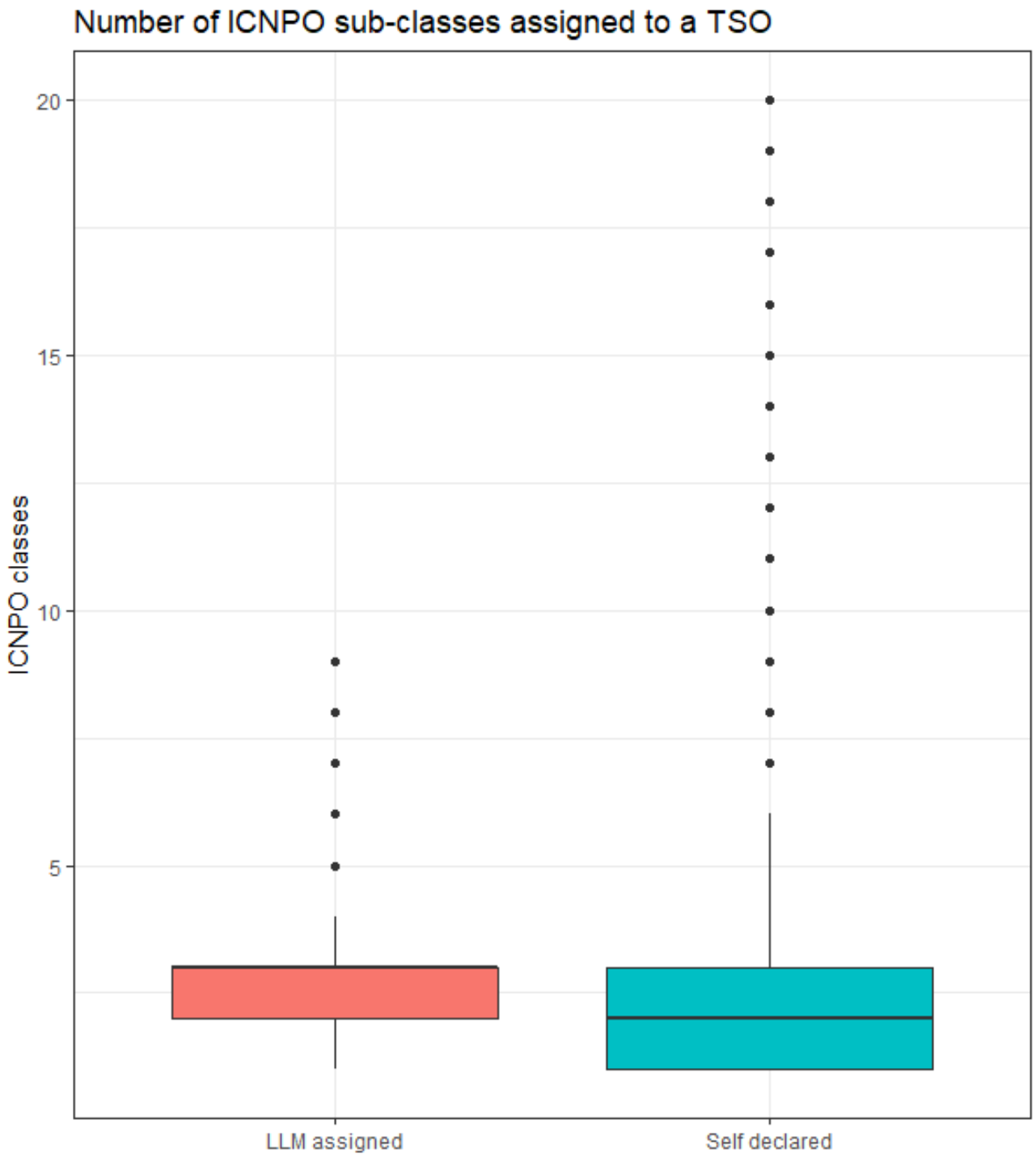
RETURN A DICTIONARY with this format:

```
{'classifications’:
  [{
    'macro_category': 'XX000’,
    'macro_category_confidence': 0.95,
    'justification': 'brief explanation of
this choice’,
    'subcategories': [{
      'code': 'XX100’,
      'confidence': 0.90,
      'justification': 'brief explanation’
    }]
  }]
}
```

TSO ECON. ACTIVITY: RESULTS (I)

Some TSOs self-declare their main economic activity (right box-plot) and we used them as a reference with which to compare the output of the LLM.

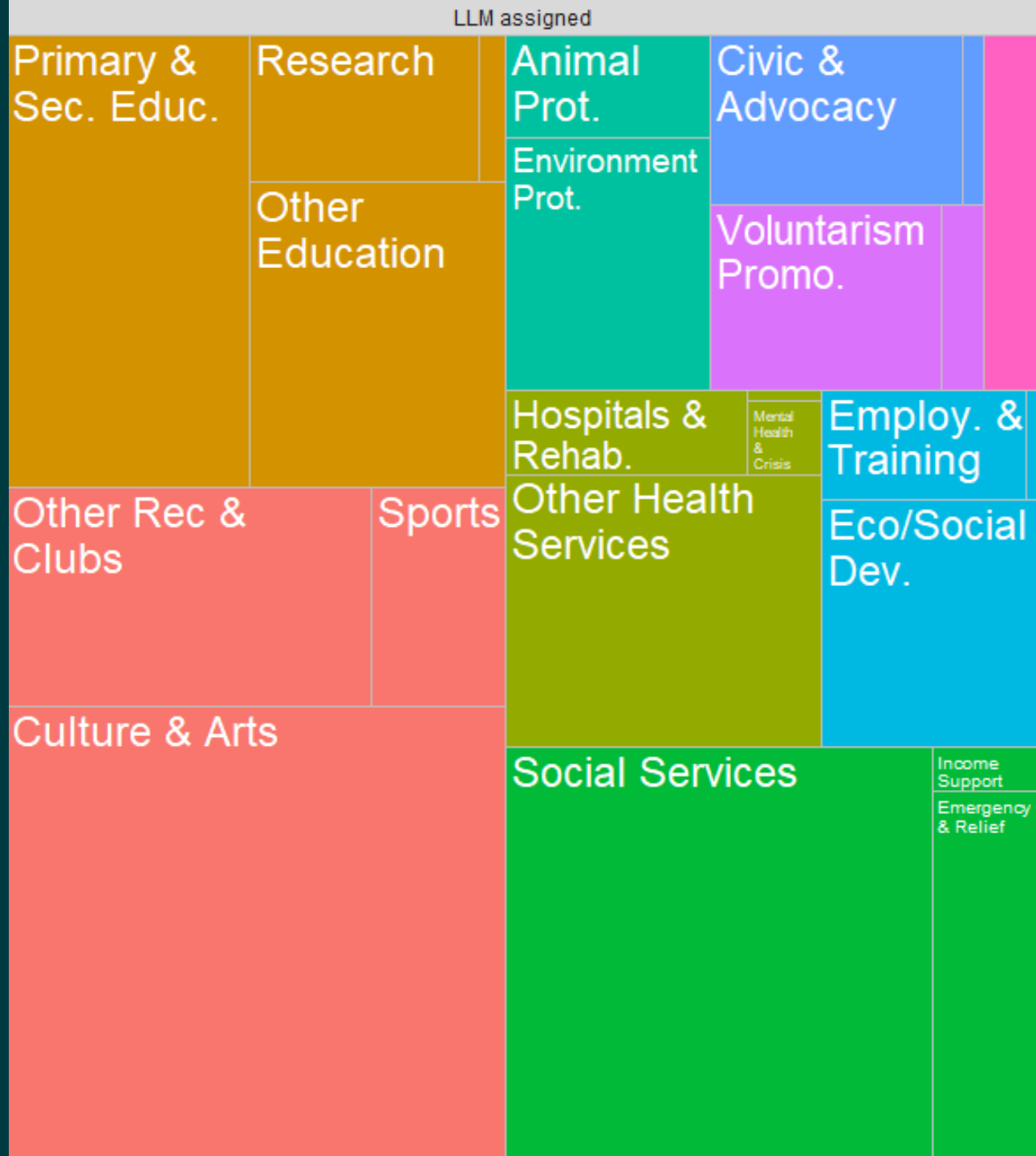
Our LLM classifier (left) tends to assign the same number of ICNPO sub-classes to a TSO (2.60 on average, compared to 2.48 self-declared sub-classes). However, we mostly avoid cases with plenty of classes assigned to a single TSO.

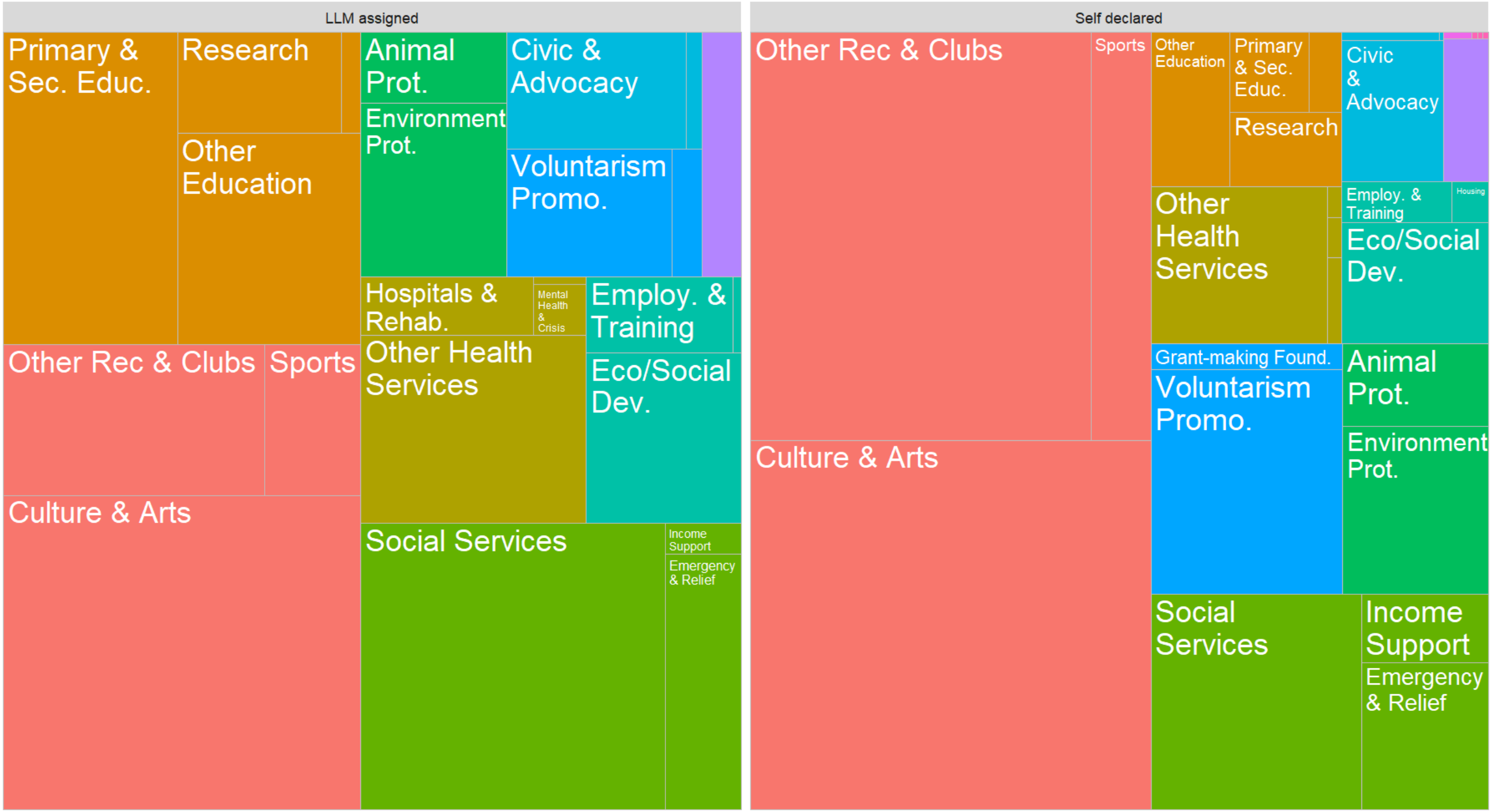


TSO ECON. ACTIVITY: RESULTS (II)

In this picture, we select one ICNPO sub-class (at random) for each TSO.

The following slide compares this picture with the corresponding one for the 88,043 TSOs that self-declared at least a ICNPO sub-class.





Considering any TSO with either one or the other classification

TSO ECON. ACTIVITY: DISCUSSION

- Our feeling, also confirmed by a few experts and practitioners, is that TSOs often declare a class to self-represent their vision and long-term goals rather than the economic activity they actually perform.
- In this respect, the use of an LLM may be helpful to be more “objective” and consistent among organizations.
- Still, further exploration and assessment of this point will be needed.

WORK IN PROGRESS & FUTURE DEVELOPMENTS

- We just finished collecting data about the whole RUNTS and we plan to do so on a quarterly basis.
- We plan to establish whether a TSO is still active based on its “web activity”.
- We are planning to extract information from TSOs' balance sheets and social reports (deposited in PDF format into the RUNTS), to:
  - Improve the “economic activity” classification.
  - Answer questionnaires using LLMs.
  - Provide estimates of the contribution of the Third Sector to the national economy.
- We are developing scientific partnerships to conduct research and to evaluate the linkage with other data sources (e.g., job activations in TSOs).

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